

CHE 2201 – Physical Chemistry II

Phase Equilibria

Lec. 5. Phase Diagrams of Non-Ideal Mixtures

5.1. Introduction

During the previous lecture, we have studied binary systems of completely miscible liquids that contain only one liquid phase. However, there are some liquid mixtures which are partially miscible (e.g. phenol and water).

When phenol is added to water, initially the substance dissolves completely. With further addition of phenol into water, two layers separate out. Of these two layers, one layer is water saturated with phenol, and the other is phenol saturated with water. These two liquid layers (or phases) in equilibrium are called conjugate solutions. The two liquids are partially miscible in one another.

Thus, the conjugate solutions means two different solutions in equilibrium, each consisting of two liquids, one liquid being the solvent in one of the solutions, and the solute in the other.

5.2. Chemical Potential and Miscibility

Usually, a solution is formed from its constituent components spontaneously, and therefore its formation at const. p and const. T , is associated with a reduction in the Gibbs Free Energy (G) of the system.

From Thermodynamics, we know that the total differential of the Gibbs free energy for an open system (Refer to first year lectures):

$$dG = V dp - S dT + \sum \mu_i dn_i \quad (5.1)$$

where $\mu_i = (\partial G / \partial n_i)_{p, T, n_j (j \neq i)}$ – chemical potential of component i .

When p and T – const.

$$dG = \sum \mu_i dn_i \quad (5.2)$$

Hence, for a binary system of two components A and B, we can write

$$dG = \mu_A dn_A + \mu_B dn_B \quad (5.3)$$

Let us derive an important relationship between the chemical potentials of a component, entering the composition of several phases of a heterogeneous system. The transition of the amount dn_i of the component i from one phase (phase 1) to another phase (phase 2) at equilibrium and at const. p and T , results in a change in the Gibbs free energy (dG).

This change (dG) consists of the changes in the Gibbs free energies of the two phases (dG^1 and dG^2).

$$dG = dG^1 + dG^2 = \mu_i^1 dn_i^1 + \mu_i^2 dn_i^2 \quad (5.4)$$

Since $dn_i^2 = -dn_i^1$, and on the condition of equilibrium $dG = 0$:

$$\mu_i^1 dn_i^1 + \mu_i^2 (-dn_i^1) = 0 \quad (5.5)$$

$$\mu_i^1 - \mu_i^2 = 0$$

$$\mu_i^1 = \mu_i^2 \quad (5.6)$$

This conclusion can obviously be extended to any number of phases.

$$\mu_i^1 = \mu_i^2 = \dots\dots = \mu_i^P \quad (5.7)$$

If we recall the derivation of the Gibbs Phase Rule (Refer section 1.2), we will realize that we have already used this relationship without derivation (see equation 1.3.b).

Equation 5.7 tells us that the chemical potentials of a given component in all phases of a system in equilibrium, are equal to one another.

Now, let us consider a system in equilibrium, which contains only two liquid phases. Let one of these liquid layers consist of pure liquid A, and the other layer is a saturated solution of A in B.

The thermodynamic requirement for equilibrium is that the chemical potential of A in two phases should equal to one another. In other words, chemical potential of A in solution (μ_A) is equal to that in the pure liquid (μ_A^0).

$$\mu_A = \mu_A^0$$

$$\text{or } (\mu_A - \mu_A^0) = 0 \quad (5.8)$$

From thermodynamics, we know that for an ideal system

$$G = G^0 + RT \ln p \quad (5.9)$$

where G^0 is the standard free energy.

Similarly:

$$\mu_i = \mu_i^0 + RT \ln p_i \quad (5.10)$$

$$\mu_i = \mu_i^0 + RT \ln C_i \quad (5.11)$$

If we substitute C_i (or p_i), by mole fraction X_i , we get

$$\mu_i = \mu_i^0 + RT \ln X_i \quad (5.12)$$

Thus, from equations (5.8) and (5.12), we get

$$(\mu_A - \mu_A^0) = RT \ln X_A \quad (5.13)$$

where X_A is the mole fraction of A in the solution.

It is clear from the equation 5.13, that $RT \ln X_A$ becomes zero, only if $X_A = 1$, i.e. if the solution consists only of A and not B.

In other words,

$$X_A = (n_A/n_A + n_B) \longrightarrow 1, \text{ only when } n_B \longrightarrow 0.$$

But, for a mixture of two liquids, $X_A \neq 1$. It will be less than 1, and $\ln X_A$ will be negative.

When $\ln X_A = -1$, $(\mu_A - \mu_A^0) = -RT$.

Similarly, $(\mu_A - \mu_A^0)$ values will be negative for all compositions of the ideal solution.

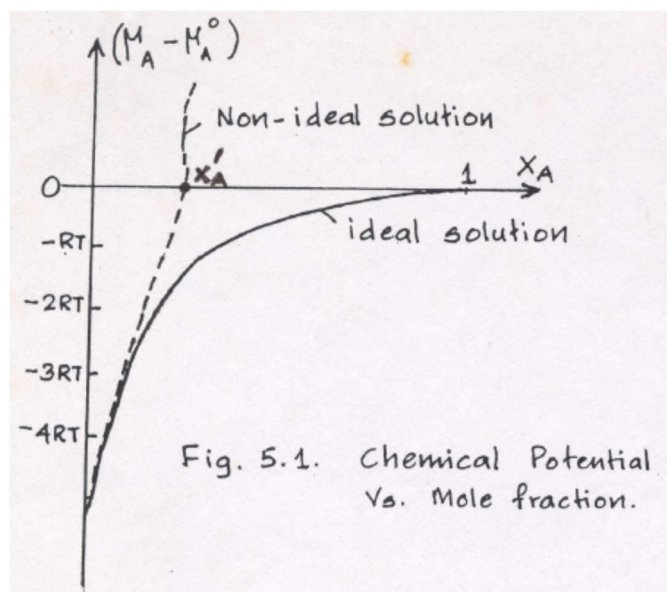


Fig. 5.1. Chemical Potential vs. Mole Fraction

The figure 5.1 shows a plot of $(\mu_A - \mu_A^0)$ against X_A , in which the values of $(\mu_A - \mu_A^0)$ are negative. It indicates that pure A can be transferred to an ideal solution with a decrease in free energy. Consequently, components A and B form completely miscible ideal solutions.

For a partially miscible mixture, $(\mu_A - \mu_A^0)$ values must be negative for some compositions of the mixture, and it reaches zero just before the point at which they form separate layers. From the compositions of O to X_A' (Fig. 5.1), the components are miscible, but at point X_A' the value $(\mu_A - \mu_A^0)$ is equal to zero. So, the system can exist as a solution having mole fraction of A = X_A , and a separate layer of pure liquid A. The value X_A' indicates the solubility of A in B in mole fraction. At this moment, A spontaneously separates from the solution into the pure liquid A, thus reducing X_A until it reaches the equilibrium value of X_A' .

It is clear from this discussion, that the solution contains X_A' mole fraction of A, and the separate layer contains only pure liquid A. In general, partially miscible liquids are considered as non-ideal solutions and corresponding curve is different from that of the ideal curve.

5.3. Phase Diagrams of Partially Miscible Liquids

Let us consider a partially miscible system in accordance with the Phase Rule. When both components A and B are miscible in each other, the behaviour of the system is similar to completely miscible liquids.

Phase Rule: $F = C + 2 - P$

No. of Components – $C = 2$

No. of Phases – $P = 2$ (one liquid and one gaseous)

$$F = 2 + 2 - 2 = 2$$

The system has two degrees of freedom, so it is bivariant.

When more B is added to the solution, a point will be reached (See Fig. 5.1), where two layers separate. Then there will be two liquid phases and one gaseous phase. The number of components will remain as two.

Then: $F = 2 + 2 - 3 = 1$. – So, the system is univariant.

Since the system is univariant, one variable (T, p or composition) should be kept constant to define the system completely. That means at a given T, the p and composition remain constant. Therefore, addition of one of the components does not change the concentration or T, but it changes the relative amount of the phases.

In practice, we keep p constant as in the previous cases. At ordinary T, with the addition of B two layers formed. If the T is raised (or in some cases lowered), the second phase will disappear, or in other words, the liquids are completely miscible at the increased T. A further addition of B at this T, will again produce two separate layers. This procedure can be used to obtain solubility curve of B in A at various Ts. Actually, the final curve is made up of two separate solubility curves. One of the curves shows the solubility of A in B, and the other shows the solubility of B in A. The two curves merge into each other at a maximum or a minimum T.

When one liquid is dissolved in the other, heat may be either evolved or absorbed. If heat is absorbed, then we have to supply heat resulting a T increase in the system. Therefore, the curve shows a maximum in such systems, whereas a minimum is observed for a system which liberates heat during the mixing.

5.3.1. Solubility Curve with a Maximum

In these systems, solubility increases with the increase of T. Examples are phenol/water, water/isobutyl alcohol. The figure 5.2 shows the solubility curve of the two components A and B with a maximum.

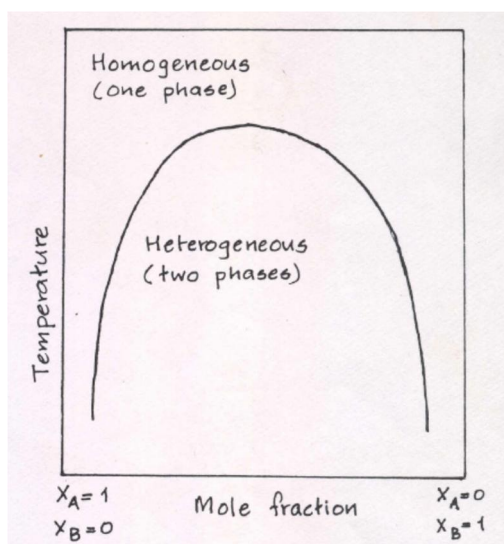


Fig. 5.2. Solubility Curve with a Maximum

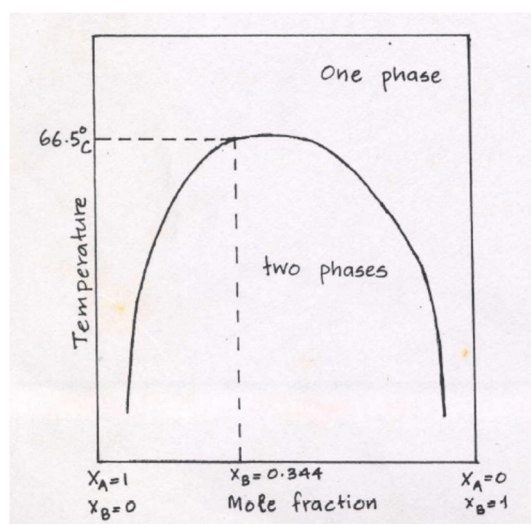


Fig. 5.3. Solubility Curve of Phenol/Water System; A – Water and B – Phenol

Let us consider the phenol/water system (Fig. 5.3). As discussed earlier, at a certain concentration, phenol appears to be insoluble in water with the formation of another layer. One layer has phenol dissolved in water, and the other has water dissolved in phenol. With the increase of T , the concentrations of phenol in the water layer, and the water in the phenol layer increase. Again at a particular T , the composition of the two layers become identical and the two liquid solutions move into one homogeneous solution.

The temperature at which the two layers (conjugate solutions) become identical is known as the critical solution temperature, and the corresponding concentration as the critical concentration. Above this critical T , the two liquids which had previously been partially miscible become completely miscible in all proportions, and then only one liquid phase is possible.

The critical temperature for phenol/water system is 66.5°C and the critical concentration is 34.4% of phenol in water. (See Fig. 5.3). Above the critical T , the two components (i.e. phenol and water) are completely miscible in each other, and form a homogeneous solution. Below this, two layers, containing phenol in water and water in phenol in different concentrations are in equilibrium with each other. Thus, the system is heterogeneous with these two layers.

The composition of the conjugate solutions can be determined by drawing a tie-line through X , parallel to the composition axis, as shown in Fig. 5.4.

The composition of one layer at X_1 , is given by x_1 , and of the other layer at X_2 is given by x_2 . As the T increases, the points X_1 and X_2 travel upwards along the boundary line to the point U_1 . At point U_1 , the last traces of X_1 and X_2 disappear, and the system becomes homogeneous.

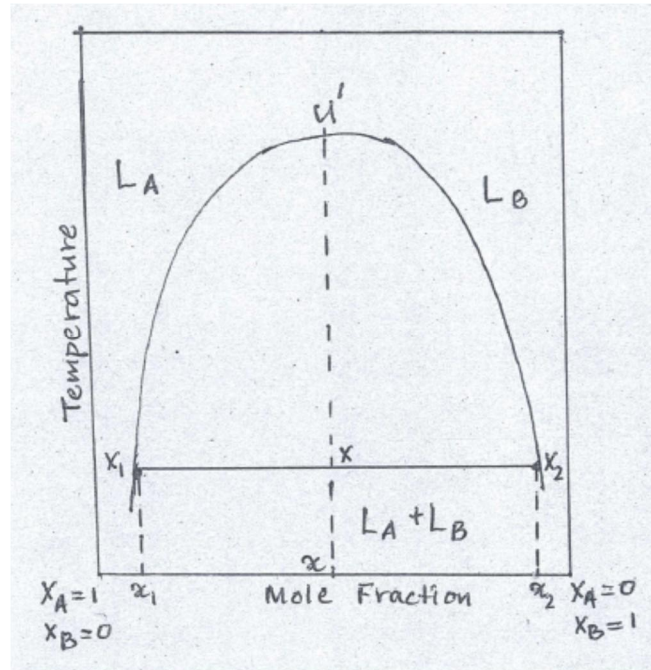


Fig. 5.4. Partially Miscible Liquids – Determination of the Composition

5.3.2. Solubility Curve with a Minimum

In some cases, the solubility of one liquid in another decreases with rise of temperature. One such example is **triethylamine and water**. When these two liquids are mixed, they form a homogeneous system at 18° C (see Fig. 5.5).

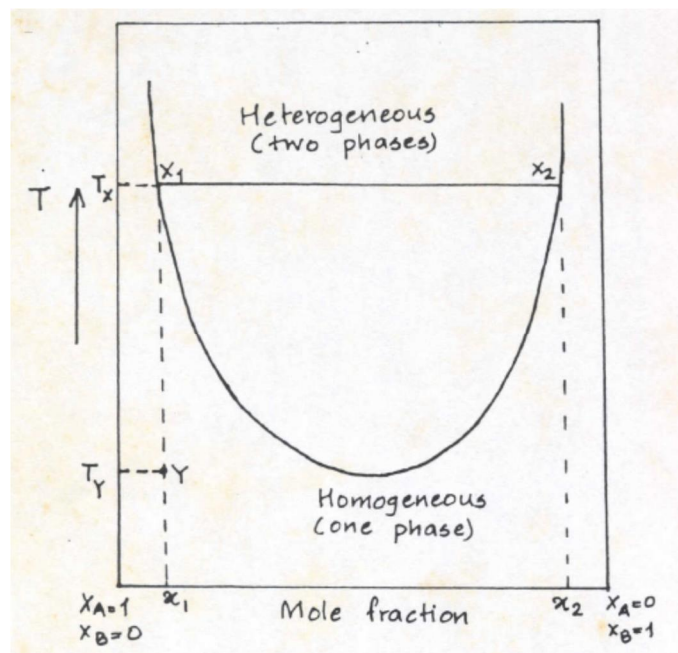


Fig. 5.5. Solubility Curve with a Minimum; A – Water; B – Triethylamine

The solubility curve shows a minimum, and the minimum T is called the lower critical solution temperature.

Let us consider a mixture of triethylamine and water defined by the point Y , at temperature T_Y . At point Y , the solution is homogeneous. As we increase the temperature upto T_x (at X_1), the homogeneous solution separates into two layers (phases), thus the solution becomes heterogeneous. The points X_1 and X_2 give the compositions of two layers. The composition of one layer at X_1 is given by x_1 , and the other layer is given by x_2 .

5.3.3. Completely Closed Solubility Curves

In some cases, partially miscible liquids show both minimum and maximum, Therefore, the solubility appears to be a closed loop.

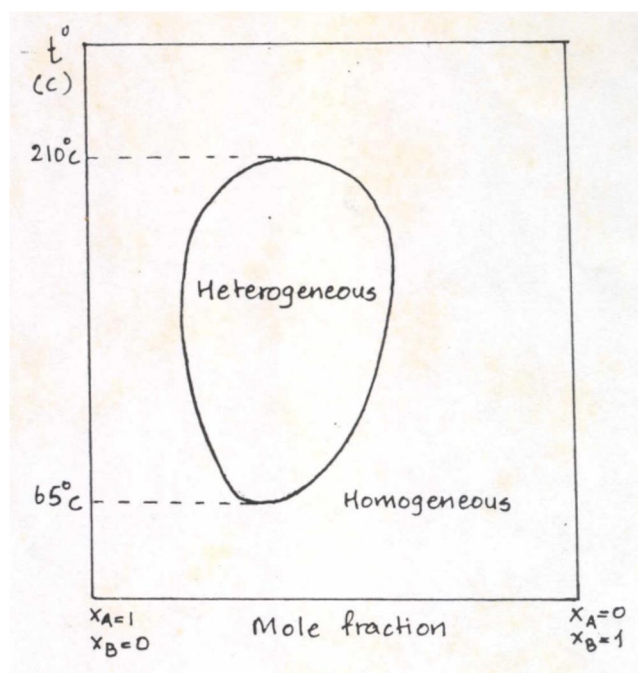


Fig. 5.6. Completely Closed Solubility Curve
A – Water; B – Nicotine ($p = 1 \text{ atm}$)

Nicotine and water are completely miscible below $61^\circ C$ and above $210^\circ C$. At all points in the closed loop two layers are present, while the points outside the loop represent homogeneous system. Closed solubility curves are shown by the systems such as, water/methyl piperidine; glycerol/benzyl ethyl amine etc.

5.4. Fractional Distillation of Non-Ideal Mixtures

Mixtures which are ideal or near-ideal solutions, can be separated into their components by fractional distillation (Refer to section 4.3). Non-ideal solutions, which do not obey the Raoult's law, do not exhibit a linear relationship between vapour pressure and mole fraction. Many systems show a positive or negative deviations in their vapour pressure curves (Refer section 3.3). This gives rise to a phase diagram that exhibits a maximum or minimum in their boiling temperatures. Such mixtures cannot be completely separated into their components by fractional distillation.

A mixture with a maximum or minimum boiling point (or vapour pressure) is called an azeotrope. (Azeotrope means boiling separately). The composition of an azeotropic mixture does not change upon distillation, and they boil at a const. temperature. (Azeotrope, like a pure liquid, boils at a constant temperature, while an ordinary mixture boils over a range of temperatures).

5.4.1. Systems with a Minimum Boiling Point

A mixture which shows positive deviation from Raoult's law, exhibits a maximum in the vapour pressure curve, and hence, the boiling point curve shows a minimum.

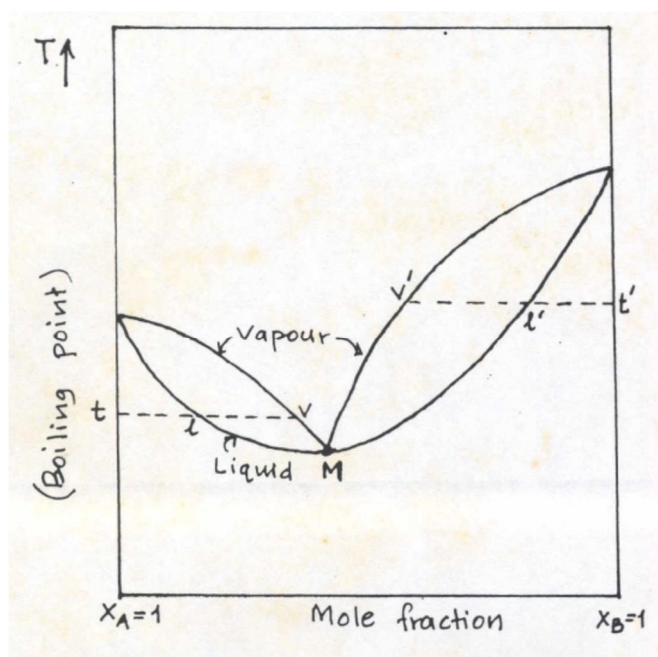


Fig. 5.7. Systems with a Minimum Boiling Point ($p = \text{const}$)

When the liquid l at temperature t , the vapour first evolved has the composition v , and is relatively richer in B than in the liquid. As distillation proceeds, the residual liquid in the flask tends towards pure A in composition. If the condensate is redistilled, however, the vapour approaches in composition that of the mixture. Ultimately, fractional distillation will result in a distillate of composition M (i.e. azeotropic composition), although the final residue will approach pure A. So, the complete separation of two liquids is not possible.

An analogous conclusion is reached by consideration of liquid l' . Here, the composition of the distillate will again tend towards M, while the residue would approach pure B.

So, we can conclude that for systems having a minimum boiling point, it is possible to obtain pure state by fractional distillation, only the component present in excess. This component will remain as the final residue in the flask. The composition of the distillate will always tend towards that of the azeotropic mixture.

We can consider these two parts as separate simple systems divided by a vertical line through the point M. Then each part takes the appearance of a simple system, in which the boiling point changes regularly with the composition. The components of the two parts are A and M (azeotropic mixture) for the left side, and B and M for the right side.

E.g.: ethanol/water; ethanol/ CCl_4 ; ethyl acetate/water; ethyl acetate/alcohol

5.4.2. Systems with a Maximum Boiling Point

A mixture which shows a negative deviation from Raoult's law, exhibits a minimum in the vapour pressure curve, and hence, the boiling point curve shows a maximum.

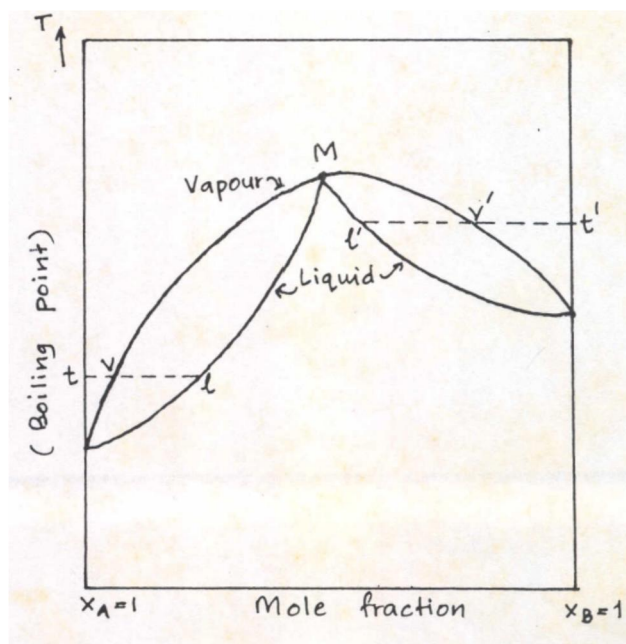


Fig. 5.8. Systems with a Maximum Boiling Point ($p = \text{const}$)

These systems are very much similar to minimum boiling point mixtures. Here too, fractional distillation gives us only the component present in excess, in pure state. However, that component will gather in the condenser as the distillate. The final residue in the flask will be of the composition of M (azeotropic mixture).

Here again, vertical line through M gives two simple systems. To the left is the system A/M, and to the right is the system B/M. The system A/M gives pure A as the distillate and M as the residue. The system B/M gives pure B as the distillate and M as the residue. E.g. water/halogen acid.

5.5. Distillation of Immiscible Liquids

Let us consider two immiscible liquids in equilibrium with vapour at a particular temperature.

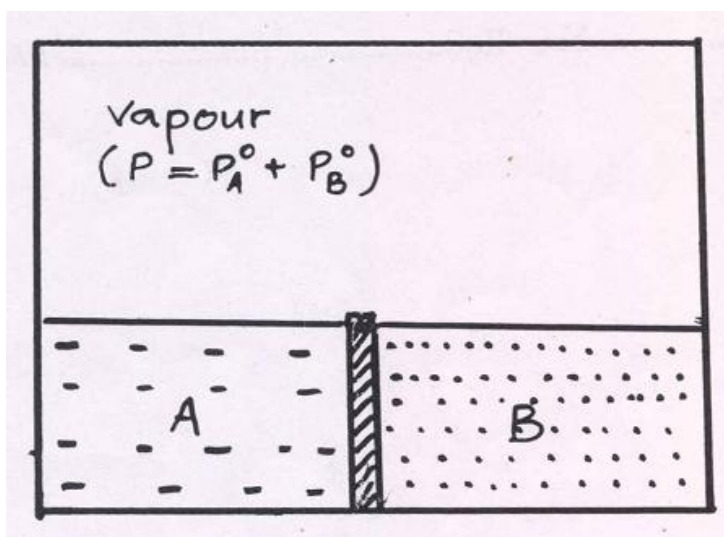


Fig. 5.9. Immiscible Liquids in Equilibrium with Vapour

The barrier between the liquids A and B only keeps the liquids apart. The total pressure is the sum of the vapour pressures of pure liquids.

$$p = p_A^0 + p_B^0 \quad (5.14)$$

The mole fractions in the vapour are:

$$X_A = p_A^0/p; X_B = p_B^0/p; \quad (5.15)$$

If n_A and n_B are the number of moles of A and B in the vapour, then

$$n_A/n_B = X_A/X_B = (p_A^0/p)/(p_B^0/p) = p_A^0/p_B^0 \quad (5.16)$$

The masses of A and B in the vapour phase are:

$$W_A = n_A M_A; W_B = n_B M_B \quad (5.17)$$

where M is the molar mass.

$$\Rightarrow W_A/W_B = (n_A M_A)/(n_B M_B) = (p_A^0 M_A)/(p_B^0 M_B) \quad (5.18)$$

The equation 5.18 relates the relative weights of two liquids present in the vapour to their molecular weights and vapour pressures. If this vapour is condensed, then the relative weights of A and B in the condensate can be calculated using the equation 5.18. The same equation can be used to study the steam distillation of liquids, e.g. natural products. Here the liquid to be distilled and water (or steam) are considered as two immiscible liquids.

In practice, steam distillation is done by passing steam through the liquid to that is to be distilled. A system of two immiscible liquids will boil, i.e. it distils freely when the total vapour pressure is equal to the atmospheric pressure. Since the total vapour pressure is greater than that of either component, the boiling point of the mixture is lower than that of either component. The vapour condensed is a mixture of the liquid (that to be separated) and water, which are immiscible. As such, they can be separated into two layers. This technique is used to separate organic compounds present in natural products.